

IRC24 (P40580) vs Nre1 (P40579): Paralog Pocket Specificity — Curator Report

Gene: IRC24 / YIR036C, *Saccharomyces cerevisiae* S288C (NCBITaxon:559292); UniProt **P40580** ("Benzil reductase ((S)-benzoin forming) IRC24", BZRD_YEAST). **Paralog:** Nre1 / YIR035C; UniProt **P40579** (uncharacterized oxidoreductase). **Hypothesis slug:** paralog-pocket-specificity-alphaFold. **Focus type:** free_text.

Executive Judgment

Verdict: Partially supported — with the two halves of the seed hypothesis landing differently.

1. "The paralogs have *diverged* in substrate preference" → **REFUTED by the structural evidence.** The AlphaFold Irc24 model and the experimental Nre1 crystal structure superpose at 1.36 Å CA RMSD, and the substrate pocket is **more conserved than the protein overall (15/18 residues identical, 83%)** with a strictly conserved catalytic Ser-Tyr-Lys triad. This "essentially shared pocket" is exactly the outcome the seed flagged as plausible given the near-identical benzil kinetics, and it is what the data show. The only hint of divergence is subtle: Irc24's pocket is slightly more hydrophobic (loss of the Nre1 Asp138 carboxylate; Irc24 65% vs Nre1 50% hydrophobic lining), a second-shell change unlikely to constitute a specificity switch.
2. "Comparing the structures *identifies the natural substrate class*" → **PARTIALLY RESOLVED (narrowed, not named).** Structure + Foldseek constrain the chemistry but cannot name a single physiological substrate. A genuine Foldseek run shows every nearest neighbour with a known substrate reduces a **ring/cyclic carbonyl** (diaryl diketone, bicyclic alkaloid, pterin, alicyclic ketone, steroid); **no aliphatic-dicarbonyl (methylglyoxal-type) reductase** appears. Of the four options the seed asked us to rank, **methylglyoxal is the least supported**, and a **flat/fused aromatic ring carbonyl (e.g., a quinone such as 1,4-naphthoquinone)** or another ring carbonyl is the best geometric fit for the large,

predominantly hydrophobic, aromatic-lined pocket. This is **inference from pocket properties + neighbour annotations, not a docking calculation**; no substrate-bound structure of either paralog exists.

Most important caveats. (a) There is **no experimental structure of Irc24 itself** — all Irc24 geometry is AlphaFold v6 (mean pLDDT 96.4; pocket pLDDT 93.6–98.8, so confidence is not the limiting factor). (b) **Neither Nre1 structure contains a substrate** (3KZV is apo + glycerol; 6UHX holds only NADP), so the pocket is inferred from the cofactor site and fold. (c) The closest *natural-substrate* homologs (sepiapterin reductase, tropinone reductase) act on substrates **yeast does not make**, so homology transfer of a specific substrate is unsafe. An honest answer is: the structure cannot upgrade the annotation from "carbonyl reductase (NADPH)" to a specific physiological substrate.

Evidence Matrix

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Con & lin
1	In-session computation (numpy NW-BLOSUM62 + Kabsch); AlphaFold DB AF-P40580-F1 v6; RCSB 6UHX	structural/computational	Refutes "diverged pocket"	Do Irc24 and Nre1 pockets differ?	53.4% global identity; 1.36 Å CA RMSD; pocket 15/18 identical ; Ser-Tyr-Lys triad conserved	Yeast SDR; AF model vs 2.75 Å crystal	High "sha pock mod Irc2 crys boun subs
2	In-session Foldseek web server (3diaa; afdb-swissprot/afdb50/pdb100)	structural/evolutionary	Qualifies substrate class	Which substrate class fits?	Neighbours = benzil reductase (6YC8 KRED1-Pglu), tropinone reductase, sepiapterin reductase, 17β-HSD14, cyclo(pent/hex)anol DH; no methylglyoxal reductase	Query = Irc24 AF v6	High neig are n carb SDR anno tran proc yeas subs
3	PMID 11796169 (Maruyama 2002)	direct assay	Supports core MF	Is YIR036C a benzil reductase?	"yeast YIR036C protein ... also reduced benzil to (S)-benzoin in vitro"; SDR related to sepiapterin reductases	<i>S. cerevisiae</i> ORF, recombinant, NADPH	Dire vitro is sy non- phys
4	PMID 33486371	direct assay + structure	Qualifies substrate class	What do benzil	"Benzil reductases are dehydrogenases	<i>Pichia glucozyma</i>	Clos char func

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	(Rabuffetti 2021)			reductases prefer?	preferentially active on aromatic 1,2- diketones"; crystal 6YC8, prefers bulky aromatic substrates	KRED1-Pglu, 1.77 Å	anal diffe orga
5	PMID 26377422 / 26952764 (Contente 2015/2016)	direct assay	Qualifies	Pocket accommodates bulky aromatics?	KRED1-Pglu "prefers space- demanding substrates"; orientation in binding pocket sets stereochemistry	Pichia benzil reductase	Supp larg hydr arom pock
6	PMID 11745140 (Maruyama 2001)	sequence/ evolutionary	Supports SDR family	Family placement	Benzil reductase is "a novel short- chain dehydrogenases/ reductase"; homology to YIR036C and sepiapterin reductases	<i>B. cereus</i> gene	Orie leve evid
7	RCSB 3KZV (apo) & 6UHX (NADP) headers; in- session HETATM scan	structural	Qualifies (negative)	Is a substrate crystallized?	3KZV: only glycerol+water; 6UHX: only NADP — no substrate in any structure	Nre1 crystals, 2.0 / 2.75 Å	Har nam subs from stru

GO Curation Implications (leads — require curator verification)

Molecular Function (MF). Evidence supports a **retained but generalized** MF. The obsoleted benzil-specific term was correctly removed (benzil is synthetic). The defensible computed/experimental MF is a **carbonyl/ketone reductase, NADPH-dependent** activity: - Candidate MF: **GO:0004090 carbonyl reductase (NADPH) activity** (or the NADP-oxidoreductase parent GO:0016616 "oxidoreductase activity, acting on the CH-OH group of donors, NAD(P) as acceptor"), with evidence code **IDA**

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in-vitro benzil reduction) for the activity and **ISS/ISA** for the structural class. The structure does **not** justify a more specific substrate term (e.g., a quinone- or steroid-specific MF) — do **not** add one as asserted. - Avoid "protein binding" — a specific oxidoreductase MF is supported.

Biological Process (BP). No BP is supported by direct evidence. The "IRC24 = increased recombination centers" gene name derives from a **loss-of-function screen phenotype**, a downstream/pleiotropic readout, not the molecular function; do not annotate a DNA-repair/recombination BP from the enzymatic data. Leave BP **unknown** / **non-core** pending a physiological substrate.

Cellular Component (CC). **Cytoplasm** (GO:0005737) is consistent with both paralogs being soluble cytoplasmic SDRs (3KZV titled a "cytoplasmic protein"); retain if already supported by localization evidence.

Paralog note for the curator: because the pocket is ~83% identical and the fold is near-identical, any substrate/function annotation transferred to one paralog should be **applied symmetrically or flagged**, and benzil-reductase annotations on either gene should carry a "non-physiological substrate" qualifier.

Mechanistic Scope

- **Direct molecular activity (what the structure tests):** NADPH-dependent reduction of a carbonyl (ketone/diketone) at the SDR catalytic centre (Tyr157/Ser143/Lys161 in Irc24), with the substrate carbonyl positioned over the nicotinamide C4 of NADP. This is the gene product's primary biochemical function.

- **Not direct / out of scope:** the "increased recombination centers" phenotype (loss-of-function, pleiotropic), any genome-stability role, and any specific physiological pathway — none are established and none follow from the enzymology.

Conflicts and Alternatives

- **Against "diverged specificity":** identical fold, 83%-identical pocket, conserved catalytic triad, indistinguishable cavity volume, and the reported near-identical benzil kinetics all point to a **shared** pocket. The seed's primary framing (divergence) is contradicted.
- **Paralog/annotation carry-over risk:** P40580 is annotated "benzil reductase" in UniProt while P40579 is "uncharacterized"; given 52% identity and a shared pocket, this asymmetry is likely annotation history, not biology. Curators should treat benzil activity as a **shared, in-vitro, non-physiological** property.
- **Homology trap:** nearest natural-substrate neighbours (sepiapterin reductase → tetrahydrobiopterin pathway; tropinone reductase → plant alkaloids; 17 β -HSD14 → steroids) have **no substrate counterpart in *S. cerevisiae***, so their annotations cannot be transferred as the yeast substrate.
- **Alternative substrate hypotheses not excluded:** a **quinone** (e.g., 1,4-naphthoquinone; fits a flat hydrophobic aromatic pocket and is chemically plausible for oxidative-stress/quinone handling) is the best geometric match among the four options, but this is **not proven**; a broader "ring carbonyl" class remains open.

Knowledge Gaps

1. **No physiological substrate identified.** Checked: structure (no bound substrate in 3KZV/6UHX), Foldseek neighbours (all exogenous or non-yeast substrates), literature (only benzil assayed for the yeast protein). Matters because GO MF specificity and any BP depend on it. Resolve with substrate screening (see below).
2. **No Irc24 experimental structure.** Checked: AlphaFold v6 only (high confidence, pLDDT ~96). Matters marginally — confidence is high — but a crystal or a substrate co-complex would settle pocket geometry and the Asp138→Gly145 effect.

3. **Functional consequence of the Nre1 Asp138 / Irc24 Gly145 difference.** Checked: it is the single non-conservative near-pocket change (4.8 Å, second shell). Matters only if the paralogs are subtly specialized. Resolve by reciprocal point-mutant kinetics.
 4. **Quantitative pocket volume/druggability.** Checked only with a crude grid (relative comparison valid, absolute unreliable); fpocket/CASTp/SiteMap were not run. Resolve with a dedicated cavity tool for a defensible volume and a docking panel (naphthoquinone vs methylglyoxal vs benzil vs a steroid).
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Discriminating Tests

1. **Substrate panel kinetics** on recombinant Irc24 **and** Nre1 in parallel: 1,4-naphthoquinone and other quinones (menadione, juglone), methylglyoxal, a model steroid/ring ketone, vs benzil — measure k_{cat}/K_M and NADPH vs NADH. This directly tests "shared vs diverged" and names the preferred class.
 2. **Co-crystal or cryo-soak** of Nre1 (or Irc24) with a candidate substrate + NADP(H) to define the real pocket.
 3. **Reciprocal mutants** Irc24 G145D and Nre1 D138G; assay for any substrate-preference shift.
 4. **Dedicated pocket/docking analysis** (fpocket/CASTp + AutoDock Vina) on the AlphaFold Irc24 model with the four candidate ligands for a computed ranking to replace the current inference.
 5. **Genetics/metabolomics:** irc24Δ nre1Δ (single and double) metabolite profiling and quinone/oxidative-stress sensitivity, to connect the enzyme to a pathway.
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Curation Leads (require curator verification)

- **Action on the review:** Treat the seed's "diverged substrate preference" claim as **not supported**; record the computed "shared pocket" result instead. Treat "identifies the natural substrate class" as **narrowed, not resolved** (ring/fused-ring carbonyl favoured; methylglyoxal disfavoured).
- **Candidate MF term:** GO:0004090 (**carbonyl reductase (NADPH) activity**) or the generalized NAD(P) CH-OH oxidoreductase parent — IDA from

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keep a "benzil = non-physiological substrate" note. Do **not** assert a quinone/steroid-specific MF.

- **BP/CC:** BP **unknown** (do not infer recombination/DNA-repair from the gene name); CC **cytoplasm** if localization-supported.
- **Candidate references + exact snippets to verify:**

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— "We isolated the genes encoding yeast YIR036C protein and gerbil sepiapterin reductase, and both recombinant proteins also reduced benzil to (S)-benzoin in vitro."

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— "Benzil reductases are dehydrogenases preferentially active on aromatic 1,2-diketones."

- **Suggested curator questions:** Is there localization evidence for P40580/P40579? Is the benzil annotation on P40580 experimental (IDA) or transferred? Should P40579 receive the same generalized carbonyl-reductase MF given the shared pocket?
- **Suggested experiments:** the substrate panel and reciprocal-mutant kinetics above.

Tools and database versions actually invoked (with honest limitations)

- **AlphaFold DB** REST API + file server, model **AF-P40580-F1 v6** (modelCreatedDate 2025-08-01; AlphaFold Monomer v2.0 pipeline; globalMetric/mean pLDDT 96.38) and **AF-P40579-F1 v6** (96.62). *Note: the seed referenced v4; the current release is v6.*
- **RCSB PDB** coordinate files **3KZV** (2.00 Å, apo) and **6UHX** (2.75 Å, +NADP).
- **In-session Python (numpy)** for parsing, Needleman–Wunsch (BLOSUM62), Kabsch superposition, pocket detection, crude cavity grid, polarity. **Biopython and Phenix were not applied to the Irc24 model:** the code-executor sandbox is ephemeral and cannot share files with the Phenix-tool environment, and Biopython is not an allowed import there. (RCSB PDBs are staged in the job `data/` dir if a curator wishes to run `phenix.molprobtity/phenix.superpose_pdbs` on the experimental structures.) The absolute cavity volume is an

overestimate from a permissive enclosure criterion and is reported only as a **relative** paralog comparison.

- **Foldseek web server** (search.foldseek.com API, mode 3diaa; databases afdB-swissprot, afdB50, pdb100) — real output reproduced in `provenance/irc24_nre1_structural_analysis.md`.
- **PubMed** via search_pubmed.

Provenance (computed metrics, full per-residue pocket table with pLDDT, and the Foldseek hit list) is saved to `provenance/irc24_nre1_structural_analysis.md`.

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